

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_132410 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 78%

CHANGE REQUEST

elimina il pannello solare di destra

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power margin reduction

Subsystem: power | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially exceeding peak power consumption limits during imaging mode.

[MINOR] ISSUE-002 — Thermal gradient margin reduction

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

The removal of the right solar panel may alter thermal distribution and reduce the thermal margin on the optical bench, potentially violating the $\Delta T \leq 15^{\circ}\text{C}$ requirement.

[MINOR] ISSUE-003 — Structural balance alteration

Subsystem: structure | Req: None | Type: reanalysis_required

Removing the right solar panel changes the mass distribution, potentially affecting structural dynamics and stability.

[MINOR] ISSUE-004 — Mechanical stress redistribution

Subsystem: mechanical | Req: None | Type: reanalysis_required

The removal of the right solar panel may redistribute mechanical stresses, requiring reanalysis and potential testing to ensure structural integrity.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The removal of the right solar panel may affect shock response, necessitating retesting to validate structural integrity.

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM

Reason: The thermal distribution may be altered, requiring a new thermal balance test to ensure compliance with ΔT limits.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	61%	WARN	ΔT 14.8°C — margin < 5%	—
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

No requirements directly invalidated.